A Multi-Module Limit-Order-Book Trading System

Team 1 — QuantiFi 2026 Abstract

Summary. Our strategy actively trades 18 of the 26 products with one dedicated module per product group, drawn from five strategy families: (i) market making around fixed or implied fair values, (ii) cross-strike implied-volatility smile relative value on the vouchers, (iii) EMA-anchored z-score mean reversion, (iv) rolling-line and swing-breakout trend following, and (v) blended fixed-centre / adaptive-anchor mean reversion. Robustness features include per-module exception isolation, a duplicate-callback guard, compact `traderData` encoding (peak ≈ 4.98 KB) and an intraday drawdown kill switch on the three trend products. On the official backtest over days −2, −1 and 0 our strategy earned 523,191.5 COINS, and every traded product finished with a positive total.

1. Introduction

Teams are ranked on PnL by the average across multiple out-of-sample runs, and on Breadth, Depth and Communication from the script and this abstract. Our design therefore aims to maximise expected total PnL by spreading it across many independent return streams, each kept simple and causal, while an intraday drawdown guard limits the damage a single adverse regime can do to the out-of-sample average.

The 26-product universe falls into five groups: fixed-fair and pearl products (`AMETHYST`, `PINK/GOLDEN/BLUE/BLACK/WHITE_PEARLS`); the underlying `LYCHEE` and ten vouchers (`VOUCHER_6800–8100`); mean-reverting products (`PEANUT_BUTTER`, `MANGOSTEEN`, `TOFFEE_CAKE`, `JELLY`); trending products (`PAPER_LANTERNS`, `TELLURIUM`); and the basket group `FRUIT_BASKET`, `DRAGONFRUIT` and `GRANADILLA`.

Untraded products. Eight products receive no orders. `LYCHEE` is the spot input S of the voucher model, and `BLUE/BLACK/WHITE_PEARLS` feed the implied fair value of `PINK_PEARLS`. `VOUCHER_6800` and `VOUCHER_8100` are the wing strikes, where a leave-one-out smile fit must extrapolate rather than interpolate, so the voucher module trades only the eight interior strikes. `FRUIT_BASKET` is traded on its own price trend (Section 2.4), so `DRAGONFRUIT` and `GRANADILLA` are left untraded at 0 lots (see future work).

2. Strategy Architecture

Table 1 maps each module to the products it trades. Parameter values live here; the subsections explain the reasoning behind each design.

Product(s)	Module (script ID / config)	Key parameters
AMETHYST	Fixed-fair maker (`A50`)	fair 10000, cap 50, clip 5
PINK_PEARLS	Implied-fair passive maker (`passive_implied_w2`)	width 2, skew 2, cap 20
GOLDEN_PEARLS	Avellaneda–Stoikov-style maker (`GP_CONFIG`, mode `mm_as`)	constant fair, skew 2, k_σ 1, min half-width 1, size 10, cap 10
VOUCHER_7000–7900 (8)	Smile relative value (`S_quad7_interior8_q50`)	thr 7.0, min_fit 4, r 0, tte_days 8 (inactive at r = 0), cap 50 each
PEANUT_BUTTER	EMA z-score MR, frozen-anchor exit (`H3_frozen_anchor_exit`)	anchor_hl 500, sd_hl 1000, warm-up 300, z 2, max hold 1000, cap 50
MANGOSTEEN	EMA z-score MR (`MANGOSTEEN__MR500_z2`)	hl 500, z 2, cap 40
PAPER_LANTERNS	Rolling-line trend with hysteresis (`P_line50_gate_q50`)	window 50, entry 0.05, exit 0.02, fair gate on, forecast horizon 50, cap 50
TELLURIUM	Swing-breakout trend (`TELLURIUM_swingbreak_plain_d10_q40`)	threshold 10, cap 40
FRUIT_BASKET	Own-price EMA-crossover trend (`H2_own_basket_trend`)	fast hl 50, slow hl 200, entry band = half-spread, warm-up 500, cap 40
TOFFEE_CAKE, JELLY	Blended MR (`TJ_CONFIG`: fixed + EMA legs)	5000 ± 40 / 2000 ± 80; EMA hl 3000, z 2; cap 25 + 25; best-level (L1) entry/exit with reservation

Table 1. Modules and key parameters (caps in lots; hl = EMA half-life in ticks).

2.1 Market making

`AMETHYST` and the pearls trade around values that are either constant or recoverable from related books, so the edge is in providing liquidity rather than predicting direction. `PINK_PEARLS` takes its fair value from the other pearls instead of its own noisy mid. `GOLDEN_PEARLS` shifts a reservation price against inventory, $r = \text{fair} - \gamma \cdot q \cdot \sigma^2 - \text{skew} \cdot q/\text{cap}$ (q = position, σ^2 = EWMA of squared mid changes), quoted at half-width $\max(1, k_\sigma \cdot \sigma)$, so quotes lean toward flattening inventory. These modules earn little but almost every day.

2.2 Voucher smile relative value

Voucher prices across strikes should be consistent with one smooth volatility smile; a strike that deviates from its neighbours is likely mispriced. Using the Black–Scholes call with $r = 0$ written in total volatility $w = \sigma\sqrt{T}$,

$$C(S, K, w) = S \cdot N(d_1) - K \cdot N(d_1 - w), \quad d_1 = [\ln(S/K) + w^2/2] / w,$$

we solve each strike's implied w_k by safeguarded Newton, then fit a least-squares quadratic in moneyness $x = \ln(K/S)$, **leaving the target strike out**:

$$\hat{w}(x) = a + b \cdot x + c \cdot x^2, \quad \text{fair}_k = C(S, K, \hat{w}_{-k}(x_k)).$$

We buy when $\text{fair}_k - \text{ask} \geq 7.0$ and sell when $\text{bid} - \text{fair}_k \geq 7.0$. The leave-one-out fit stops a mispriced strike from pulling the smile toward itself. A strike is fitted only if at least `min_fit` = 4 other strikes have a valid implied volatility; otherwise it is not traded that tick. Because the rate is set to $r = 0$, the price depends on T only through $w = \sigma\sqrt{T}$, so the configured `tte_days` = 8 does not enter the live calculation (it is used only when a non-zero rate is set). The model therefore needs only current quotes.

2.3 EMA z-score mean reversion

`PEANUT_BUTTER` and `MANGOSTEEN` oscillate around a slowly moving level. We measure deviation from a lagged EMA anchor and scale it by an EWMA volatility (for `PEANUT_BUTTER`, anchor half-life 500 ticks and volatility half-life 1000 ticks):

$$d_t = \text{mid}_t - \text{EMA}_{t-1}(\text{mid}), \quad z_t = d_t / \sqrt{\text{EWMA}(d^2)}, \quad \text{enter when } |z_t| \geq 2.$$

For PEANUT_BUTTER the exit target is the anchor frozen at entry, so a drifting EMA cannot move the take-profit level away from the trade, and a maximum holding time caps exposure if reversion fails.

2.4 Trend following

PAPER_LANTERNS, TELLURIUM and FRUIT_BASKET show persistent intraday drifts that mean-reversion modules would fight. PAPER_LANTERNS uses the slope of a rolling linear fit with separate entry and exit bands (hysteresis), which avoids flipping in and out around a single threshold. A fair gate filters entries: the forecast exit level is the fitted value now plus 50 ticks of slope, and a level is taken only if this forecast beats the price by more than half the current spread. TELLURIUM enters on a break of the last confirmed swing high or low, using only pivots that are already confirmed, so no future data is used. FRUIT_BASKET trades its own mid with an EMA crossover: $d_t = \text{EMA}_{50} - \text{EMA}_{200}$ (half-lives in ticks, both from prior ticks). After a 500-tick warm-up it targets +40 lots when d_t exceeds half the bid–ask spread and −40 when it is below minus half the spread, and goes flat once d_t changes sign against the position.

2.5 Blended mean reversion

TOFFEE_CAKE and JELLY revert around stable centres on the observed days, but a fixed centre is risky if the level shifts out of sample. Each position cap is split into two 25-lot legs: a fixed-centre leg that trades the band 5000 ± 40 (TOFFEE_CAKE) or 2000 ± 80 (JELLY), and an adaptive leg that trades |z| ≥ 2 around a slow EMA anchor (half-life 3000). Both legs trade only at the best bid or ask (L1), net of a reservation for orders already working. If the level drifts, the adaptive leg keeps working while the fixed leg's exposure is capped at half.

3. Risk Management and Robustness

3.1 Intraday drawdown kill switch

The kill switch covers the three trend products, which carry the largest directional risk. For each product p it tracks intraday PnL and its running peak, where cash_t is accumulated from our own fills (the PnL field in the trading state is not populated during a run) and positions are marked at the mid:

$$\text{PnL}_t = \text{cash}_t + \text{pos}_t \times \text{mid}_t, \quad \text{peak}_t = \max_{s \leq t} \text{PnL}_s, \quad \text{trigger if } \text{peak}_t - \text{PnL}_t > \text{KILL_DD}_p.$$

KILL_DD is 1.5× the largest intraday drawdown observed on days −2 and −1 (20,275.5 / 40,680 / 765 COINS for TELLURIUM / FRUIT_BASKET / PAPER_LANTERNS). Once triggered, the product is flattened by crossing the touch and receives no further orders that day. Orders are remembered for 5 ticks, and the flatten size is

$$\text{qty}_{\text{flatten}} = |\text{pos}_t| - (\text{same-side orders still working}),$$

so a flatten never pushes the position through zero.

3.2 Engineering safeguards

- **Position limits:** every module has a hard cap (10–50 lots).
- **Duplicate-callback guard:** a repeated call at the same timestamp does not update rolling state twice.

- **Compact state:** rolling buffers and kill-switch state are encoded in `traderData` (peak ≈ 4.98 KB).
- **Isolation:** each module runs in its own `try/except`, so a failure in one product does not stop the others.

3.3 Out-of-sample considerations

All parameters were chosen on the three available days, so the results below are in-sample. We have no out-of-sample PnL to report, so we rely on design choices instead. Signals are causal and mostly scale-free (z-scores, smile residuals, slopes). Fixed levels are paired with adaptive ones in the blended module. Return comes from five families rather than one. The main exposures are the fixed `KILL_DD` thresholds and centres (AMETHYST 10000, TOFFEE_CAKE 5000, JELLY 2000), which may trigger too early or too late if volatility or price levels change.

4. Results

On the official backtest our strategy earned 225,130 (day −2), 188,395 (day −1) and 109,666.5 (day 0) COINS, a total of **523,191.5 COINS**. Per-product PnL is the day-over-day difference of the cumulative `profit_and_loss` field in the backtest log.

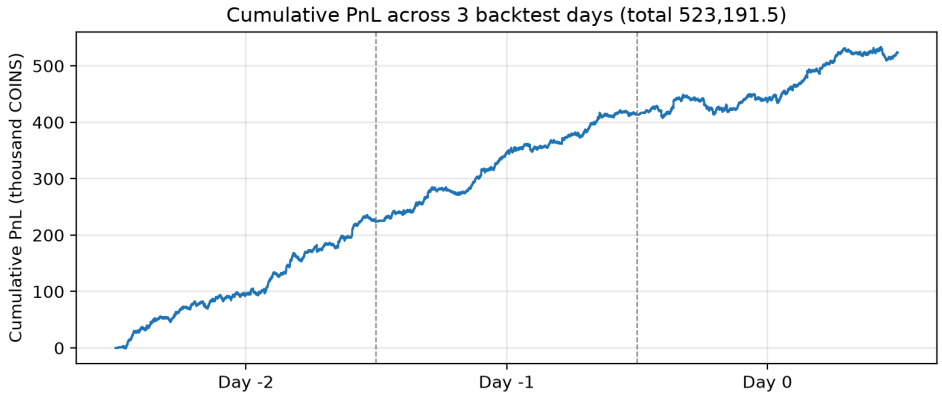


Figure 1. Cumulative total PnL (COINS) over the three backtest days.

Product	Day −2	Day −1	Day 0	Total
TELLURIUM	59,580	27,275	8,564	95,419
FRUIT_BASKET	21,419	46,734	7,703	75,856
PEANUT_BUTTER	31,292	26,615	14,671	72,578
JELLY	26,562	19,077	9,431	55,070
TOFFEE_CAKE	12,501	18,449	18,182	49,132
PAPER_LANTERNS	15,223	17,417	13,617	46,257
VOUCHER_7400	9,248	10,102	7,922	27,272
VOUCHER_7300	12,495	2,824	7,717	23,036
VOUCHER_7500	10,406	7,843	4,024	22,273
VOUCHER_7600	9,657	4,256	3,438	17,351
VOUCHER_7750	4,437	2,758	2,065	9,260
VOUCHER_7000	3,534	130	5,308	8,972
VOUCHER_7200	4,129	1,732	2,160	8,021
VOUCHER_7900	2,286	653	1,005	3,944
MANGOSTEEN	−538	1,092	3,127	3,681
PINK_PEARLS	2,289	429	119	2,837
AMETHYST	527	608	477.5	1,612.5
GOLDEN_PEARLS	83	401	136	620
Total (8 untraded = 0)	**225,130**	**188,395**	**109,666.5**	**523,191.5**

Table 2. PnL (COINS) by product and day.

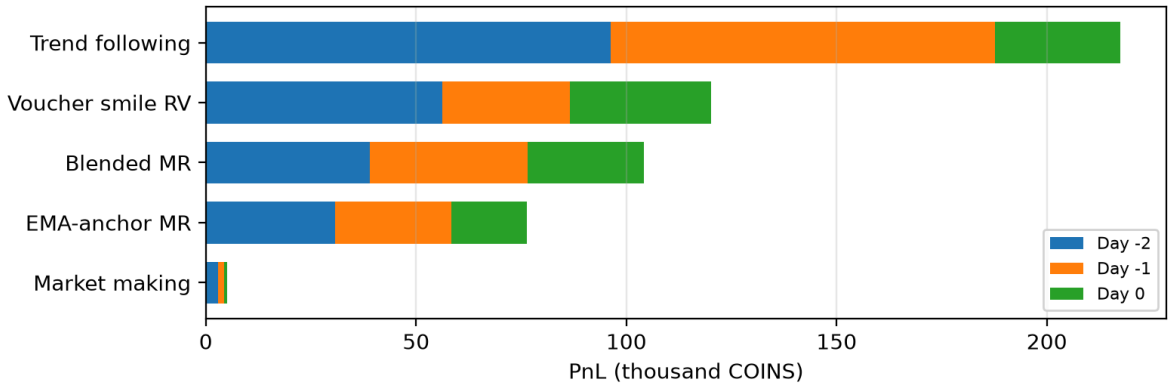


Figure 2. PnL (COINS) by strategy family and day. Totals: trend 217,532 (41.6%), voucher smile RV 120,129 (23.0%), blended MR 104,202 (19.9%), EMA-anchor MR 76,259 (14.6%), market making 5,069.5 (1.0%).

- **Breadth of contribution:** all 18 traded products have positive totals; 53 of 54 product-days are positive (the only loss is MANGOSTEEN on day −2, −538 COINS).
- **Concentration:** trend following contributes 41.6% of PnL, which is why the kill switch is applied to exactly those three products.
- **Decay across days:** daily PnL falls from day −2 to day 0, mainly in the trend family (Figure 2), from 96,222 to 29,884 COINS; the other families decline less steeply.

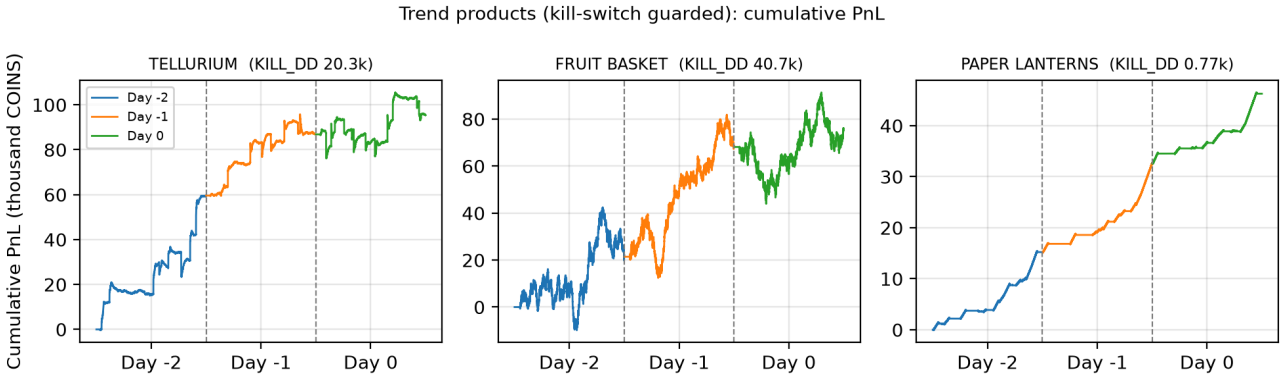


Figure 3. Cumulative PnL of the three kill-switch products across the three days, with each product's KILL_DD threshold (COINS).

5. Conclusion and Future Work

Our strategy combines product-specific modules from five families with explicit, causal risk controls, and earned 523,191.5 COINS on the official backtest. Next steps: scale `KILL_DD` and the fixed bands to rolling volatility; delta-hedge voucher inventory with `LYCHEE` and evaluate the wing strikes; and test basket-versus-component and cross-pearl trades in the untraded products.